Handbook of Biometrics: Spoof Detection

Adaptive Normalized Representation Learning for Generalizable Face Anti-Spoofing

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Contents

1. Introduction
2. Literature review
3. Paper methodology
4. Technical evaluation
5. Personal insights: extensibility
6. References

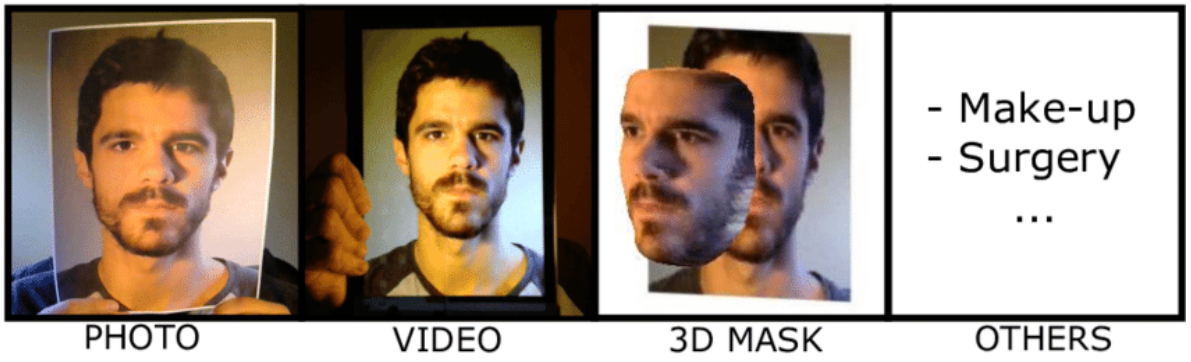
Introduction

Introduction to Face anti-spoofing

Face anti-spoofing: a security technology that determines whether a facial image or video is from a real, living person or a "spoof" attempt.

Attack Types:

- Print Attacks (Early 2000s)
- Replay (Video) Attacks (2010–2013)
- 3D Mask Attacks (2011–2016)
- Deepfake and AI-Generated Attacks (2018–Present)



Handbook core techniques and algorithms

Traditional pipeline:

1. Preprocessing
   - Face detection (or alignment)
   - Normalization
2. Feature extraction
3. Feature vector construction
4. Classification (Traditional machine learning)

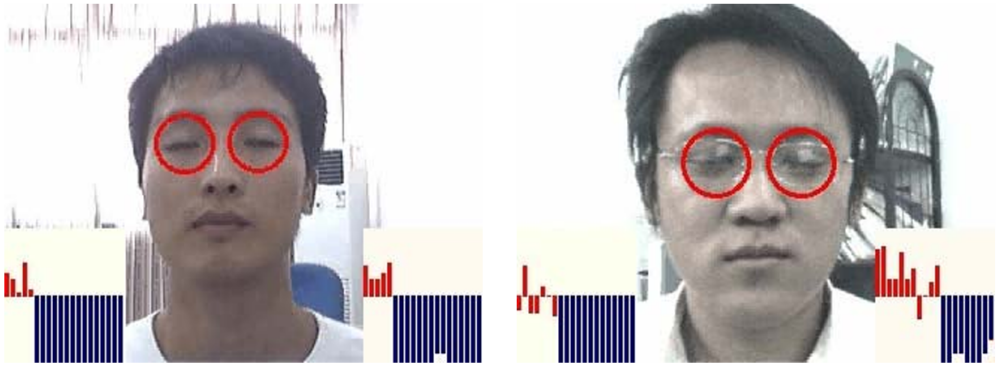
Modern pipeline - Deep learning:

1. Preprocessing
   - Face detection (or alignment)
   - Normalization
2. Deep learning feature extraction: CNN-based, Transformer, ...
3. Classification

Handbook core techniques and algorithms

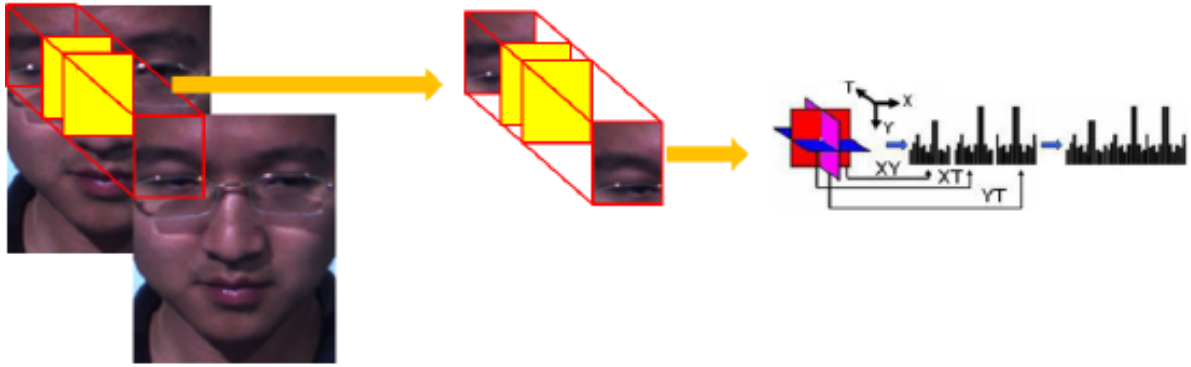
1. Motion detection
   - Eye blinking detection:
     - Region of interest (ROI)
     - Linear-chain CRF
   - Facial micro-movement analysis:
     - LBP-TOP
     - Optical flow
2. Skin texture and micro-pattern analysis:
   - LBP
   - Gabor filter
3. Thermal imaging
   - LBP (texture features)
   - Thermal gradient feature
4. Frequency-domain (Fourier) analysis: DFT
5. Challenge-response prompts

1. **Motion detection**
   - **Eye blinking detection**
     1. Detect face and extract eye ROI for each frame.
     2. Compute Eye Closity c_t from eye appearance features.
     3. Construct a Linear-chain CRF and perform sequence inference (Viterbi) to estimate eye states, detect valid blink transition patterns (Emission score and Transition score).
     4. Decide liveness based on blink occurrence.



Handbook core techniques and algorithms

1. **Motion detection**
   - **Facial micro-movement analysis**
     1. Face tracking, alignment and Region of Interest (ROI) extraction
     2. Spatiotemporal feature computation (Dynamic Texture / LBP-TOP)
     3. Optical flow based micro motion signature
     4. Temporal discriminative modeling

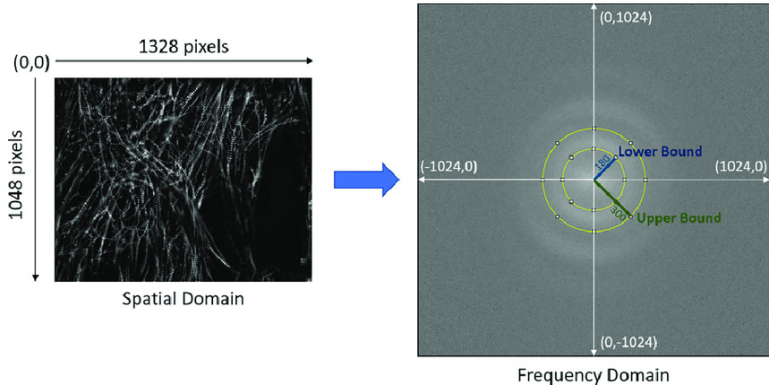


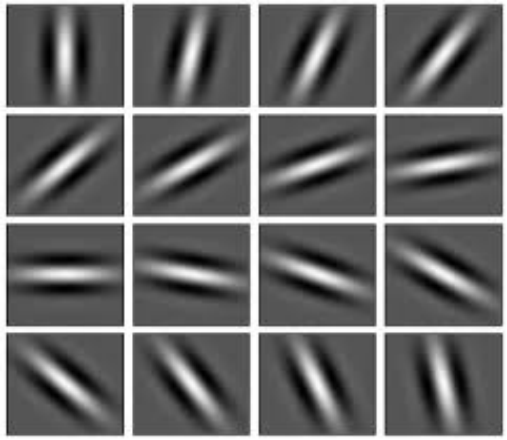
(a) (b) (c)

Handbook core techniques and algorithms

2. Skin texture and micro-pattern analysis

1. Face Region dection.
2. Local texture feature extraction (LBP)
3. Frequency-Domain Texture Analysis (DFT - Discrete Fourier Transform)
4. Multi-Scale/Orientation Filters (Gabor filter)
5. Feature Vector Assembly
6. Classification

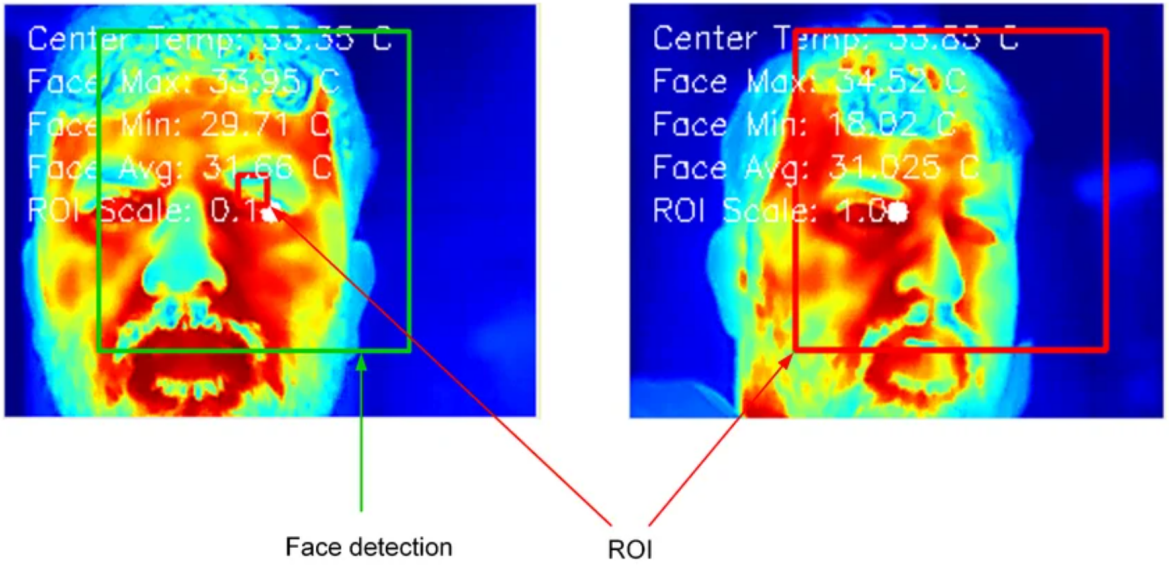




Handbook core techniques and algorithms

3. Thermal imaging

1. Acquire thermal image I_T; convert radiance to **temperature map** $T(x, y)$.
2. Detect ROI and normalize temperature
3. Extract texture (LBP), thermal gradient, and statistical features from ROI.
4. Train SVM/Bayesian classifier on feature vectors.



Literature review

History survey

A brief history of Face Anti-Spoofing techniques:

- 2000–2015: Handcrafted texture and motion-based methods.
- 2015–2018: CNN-based binary classifiers.
- 2018–2020: Auxiliary supervision (depth, rPPG).
- 2019–2023: Domain generalization and meta-learning.
- 2022–Present: Transformer-based, multimodal, and self-supervised approaches.

RGB-based Face Anti-Spoofing Methods (Chronological)

Year	Method	Core Idea	Limitation
2004	Frequency-based motion descriptor (FDD)	Temporal frequency energy from motion	Illumination sensitive; fails on replay
2005	Optical Flow Lines (OFL)	Compare inner/outer facial motion (3D vs planar)	Weak on replay; lighting sensitive
2007	Eye-blinking (CRF)	Model open/close eye sequence	Vulnerable to replay/cut-photo
2007–2009	Interactive motion	Challenge–response (blink, rotate)	Intrusive; bypassed by prerecorded video
2010–2013	Handcrafted texture	LBP, HOG, IQA micro-texture analysis	Poor generalization; weak vs 3D mask
2013–2015	LBP-TOP	Spatio-temporal dynamic texture	Higher cost; weak vs realistic mask
2016	rPPG	RGB blood-flow periodicity detection	Struggles with high-quality replay
2016–2018	CNN (spatial)	Deep spatial feature learning	Overfitting; weak cross-dataset
2018–2019	CNN + LSTM	Joint spatial-temporal modeling	High computation; unseen attack issue
2019–2021	Pseudo-depth	Depth supervision for planar detection	Limited vs realistic 3D mask
2020–2023	**Generalization methods**	**Domain adaptation, few-shot**	**Still weak vs 3D mask, DeepFake**
2022–now	Transformer / Multimodal	ViT, cross-modal, SSL pretraining	Challenged by high-quality 3D mask

Related works

- Face Anti-Spoofing methods and techniques
- Local Binary Pattern (LBP)
- LBP-TOP (Spatio-temporal LBP)
- Histogram of Oriented Gradients (HOG)
- Scale–Invariant Feature Transform (SIFT)
- CNN-based Binary Classification
- Depth-based Auxiliary Supervision
- rPPG-based Methods

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Related works

Meta-learning: Simulate domain shift during training to improve generalization.

1. Split source domains into meta-train and meta-test.
2. Update model on meta-train domains.
3. Optimize model to perform well on meta-test domains.
4. Iterate to learn domain-robust parameters.

Batch-Instance Normalization (BIN): Learn a channel-wise balance between BN and IN outputs to adaptively trade off discrimination and invariance.

- Compute normalized outputs:

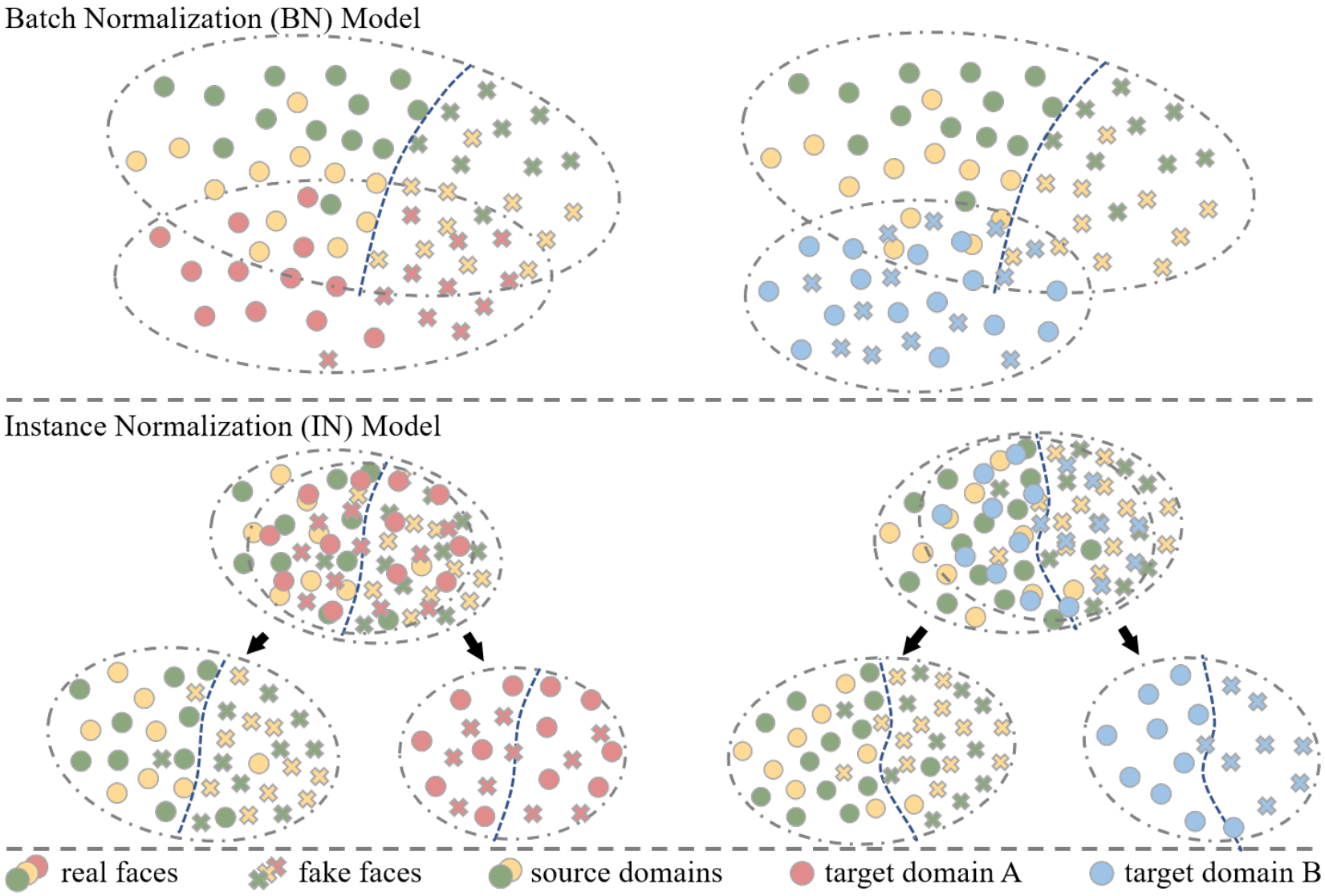
$$X_{BN} = \mathrm{BN}(X), \quad X_{IN} = \mathrm{IN}(X)$$

- Learn channel-wise parameter $\alpha \in [0, 1]$. Fuse:

$$Y = \alpha X_{BN} + (1 - \alpha) X_{IN}$$

Paper methodology

Batch vs Instance Normalization



Objective: Domains shift

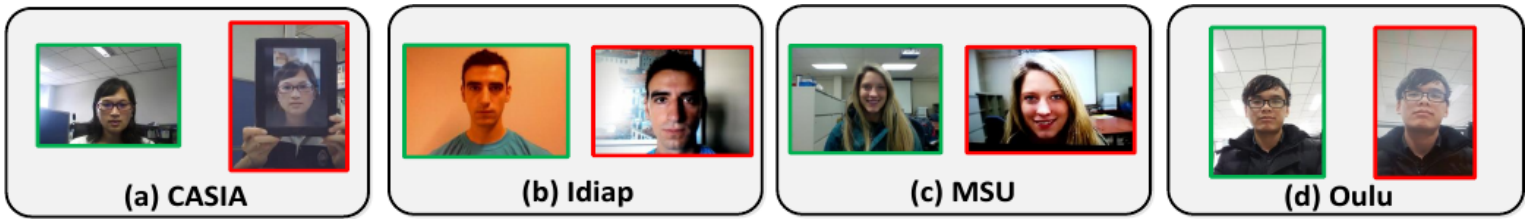
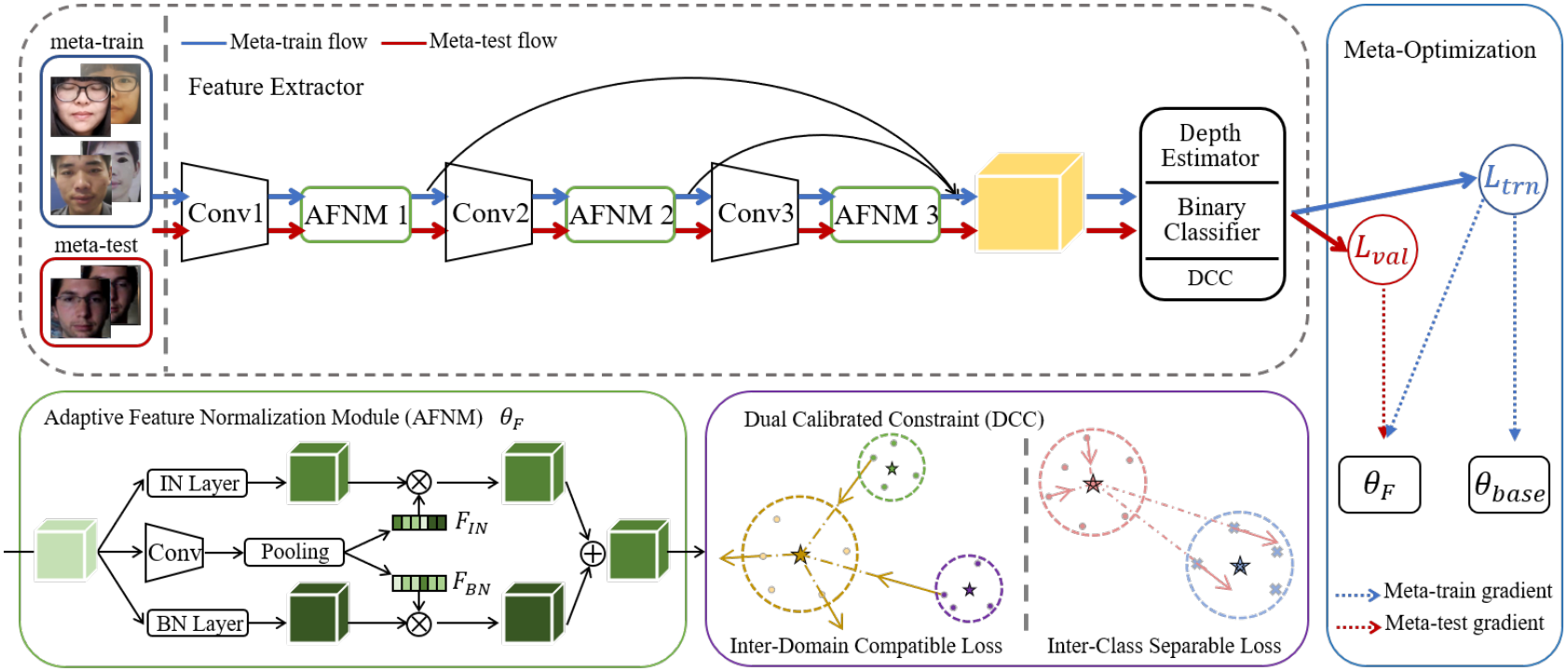


Figure: Samples from CASIA-MFSD, Idiap, MSU, and Oulu datasets exhibiting severe domain shifts.

- The framework generalizes well across previously unseen domains by remaining invariant to illumination and background variations.
- **Asymmetric class assumption**:
  - *Live*: Compact physiological similarity.
  - *Spoofs*: Highly heterogeneous.

ANRL framework overview



Adaptive feature normalization module (AFNM)

Dual calibrated constraints (DCC):

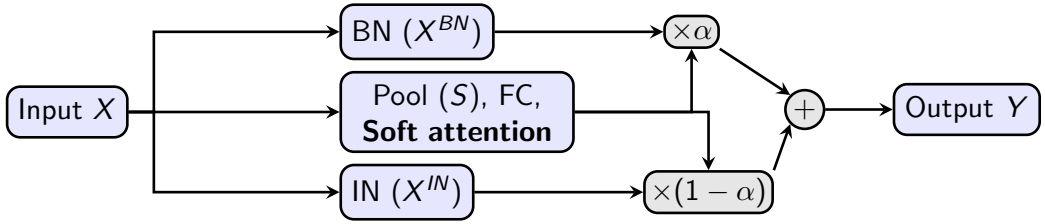
- **IDC**: Minimizes domain bias by mixing source domains. $\mathcal{L}_{IDC} = \sum_{k=1}^{K}(D_{dd}^k - D_{sd}^k)$
- **ICS**: Compacts real faces, scatters fakes. $\mathcal{L}_{ICS} = D_{rr} - D_{rf} - D_{fr}$

Note:

- IDC: reduce BN corruption due to domain bias
- D_{ff} is not included because presentation attacks are diverse.

AFNM mechanism:

- **Dual normalizations**: BN and IN applied to input X.
- **Attention factors** α_c: Computed via pooling and FC layers over X.
- **Adaptive fusion**: $Y_c = \alpha_c X_c^{BN} + (1 - \alpha_c) X_c^{IN}$.



AFNM: In-depth analysis

Core intuition:

- **BN** (*Batch Normalization*): Preserves discriminative cues but is fragile to domain shifts.
- **IN** (*Instance Normalization*): Eliminates domain style but may discard attack artifacts.

Mathematical formulation:

- **Statistic extraction**: Global average pooling captures spatial information:

$$S_c = F_{gp}(X) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} X_c(i,j)$$

- **Fully-connected layer**: A fully connected layer learns a compact representation Z:

$$Z = F_{fc}(S) = \delta(WS)$$

- **Adaptive fusion**: Soft attention derives sample-wise channel weights α_c:

$$\alpha_c = \frac{\sigma(W_B Z)_c}{\sigma(W_B Z)_c + \sigma(W_I Z)_c}, \quad Y_c = \alpha_c X_c^{BN} + (1 - \alpha_c) X_c^{IN}$$

Base model

3 primary components: a Feature Extractor (Ext), a Binary Classifier (BC), and an auxiliary Depth Estimator (Dep).

1. **Binary classification ($\mathcal{L}_{Cls}$):**
   - Optimized using standard cross-entropy loss:

$$\mathcal{L}_{Cls} = -\sum_{(x_i, y_i)} y_i \log(BC(Ext(x_i)))$$

   - Solely optimizing $\mathcal{L}_{Cls}$ often causes overfitting to domain-specific 2D textures.

2. **Depth estimation ($\mathcal{L}_{Dep}$):**
   - Prevents converging on superficial 2D artifacts. Live faces possess 3D depth, whereas spoofing attacks are flat.

$$\mathcal{L}_{Dep} = \sum_{(x_i, dep_i)} \|Dep(Ext(x_i)) - dep_i\|_2^2$$

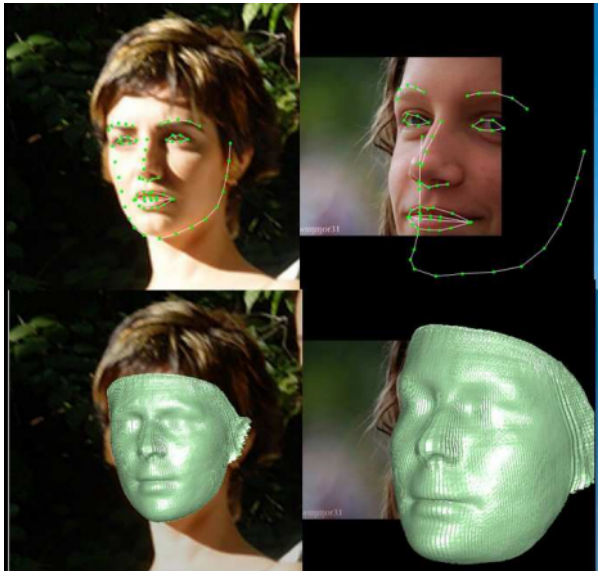


Figure: Pseudo-depth map dep_i: 3D map for live faces via PRNet [4], flat zero-map for spoofs.

Optimization strategy: Meta learning

Algorithm 1 The optimization strategy of ANRL

Require: N source domains $\mathcal{D} = [D_1, D_2, \ldots, D_N]$

Initialize parameters θ_F of AFNM and parameters θ_{base} of other modules.
Determine learning rates β_1, β_2 and hyper-parameters λ_1, λ_2.
Shuffle all samples from different domains.
for $t = 1$ to N_{epoch} **do**
    Normal train: Sample batch in $\mathcal{D}$.
    $\mathcal{L}_{base}(\theta_{base}, \theta_F) = \sum_{\mathcal{D}} \mathcal{L}_{Cls} + \mathcal{L}_{Dep}$.
    $\theta_{base} \leftarrow \theta_{base} - \beta_1 \nabla_{\theta_{base}} \mathcal{L}_{base}(\theta_{base}, \theta_F)$.
    Meta-train: Sample batch in D_{trn}.
    $\mathcal{L}_{trn}(\theta_{base}, \theta_F) = \mathcal{L}_{Cls(D_{trn})} + \mathcal{L}_{Dep(D_{trn})} + \lambda_1 \mathcal{L}_{IDC(D_{trn})} + \lambda_2 \mathcal{L}_{ICS(D_{trn})}$.
    $\theta'_F = \theta_F - \beta_1 \nabla_{\theta_F} \mathcal{L}_{trn}(\theta_{base}, \theta_F)$.
    Meta-test: Sample batch in D_{val}.
    $\mathcal{L}_{val}(\theta_{base}, \theta'_F) = \mathcal{L}_{Cls(D_{val})} + \mathcal{L}_{Dep(D_{val})} + \lambda_1 \mathcal{L}_{IDC(D_{val})} + \lambda_2 \mathcal{L}_{ICS(D_{val})}$.
    Meta-optimization: Update AFNM using combined gradients:
    $\theta_F \leftarrow \theta_F - \beta_2 \nabla_{\theta_F} (\mathcal{L}_{trn}(\theta_{base}, \theta_F) + \mathcal{L}_{val}(\theta_{base}, \theta'_F))$.
end for
return Model parameters θ_F and θ_{base}.

Technical evaluation

Technical evaluation: Strengths

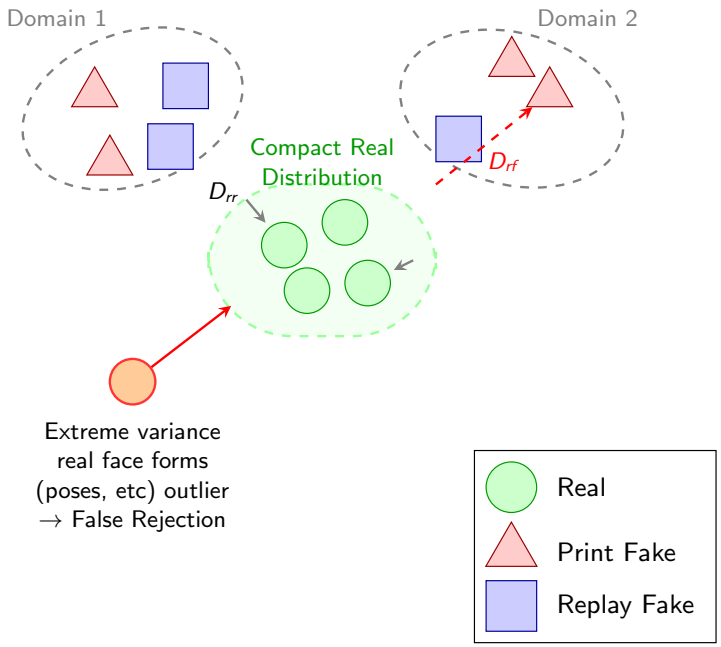
Strengths:

- **Study effects of normalization**: Previous domain generalization techniques rely on static hybrid normalizations.
- **Asymmetric class constraints for spoof detection** Compared to clustering all instances of a specific class tightly around a centroid, the ANRL framework correctly assumes that while live faces share a common physiological distribution.
- **Meta-learning optimization strategy**: By dividing the available source domains, AFNM parameters (θ_F) is forced to generalize.

Technical evaluation: Limitations

Limitations:

- **Dataset dependency**: Episodic meta-learning requires massive and highly diverse source datasets.
- **Spatial-only modality**: Ignores temporal cues, vulnerable to video replays.
- **Auxiliary supervision**: The method heavily relies on an auxiliary Depth Estimator.
- **Compactness assumption**: Genuine faces with extreme structural variance (e.g., severe poses) can fall out of D_{rr}, causing false rejections.



Technical evaluation: Applicability

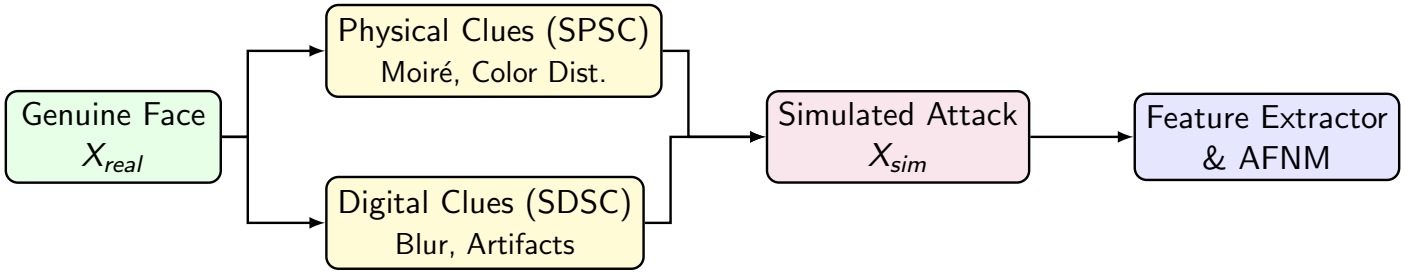
- Easy to implement: PyTorch, architecture: RFM [5].
- Security-critical deployments: environments where lighting and sensors varied, such as mobile banking authentication, etc.

Personal insights: extensibility

Future extensibility

Extension 1: Simulating Spoofing Clues (Augmentation) [3]. To overcome reliance on diverse source domains, ANRL can be extended with Joint Physical-Digital augmentation. During Meta-Train, live faces are augmented into simulated attacks (X_{sim}) using Simulated Physical (e.g., color distortion, θ_{SPSC}) and Digital (e.g., Gaussian blur, θ_{SDSC}) Spoofing Clues.

$$X_{sim} = Aug(X_{real}; \theta_{SPSC}, \theta_{SDSC}) \tag{1}$$



Future extensibility (cont.)

Extension 2: Temporal-Spectral ANRL (TS-ANRL).
While ANRL excels in 2D spatial generalization, practical FAS heavily relies on temporal cues (e.g., eye blinking). Extending AFNM with 3D convolutions allows temporal-IN to filter dynamic lighting fluctuations, while temporal-BN preserves chronological inconsistencies inherent to video replay attacks.

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